A MORPHISM OF SIMPLICIAL SPACES INDUCES A MORPHISM OF THE ZARISKI SIMPLICIAL SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the phrase 'It is clear that u is a continuous functor' (source line 375) is definitional: u is the componentwise preimage functor between open-partition topologies, for which continuity is immediate from the definition of coverings; the cofilteredness checks are carried out in full for all three conditions.

Editorial note. The label lemma-simplicial-space-site-functorial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Continuous functors and the associated adjoint pair. Sites chapter, Section \label{sites-section-morphism-sites}. Lines 375-379: f_{Zar,*} = u^s and f_{Zar}^{-1} = u_s for the continuous u.
- Exactness criterion for u_s via directed categories. Sites chapter, \label{sites-lemma-directed-morphism} and Section \label{sites-section-functoriality-PSh}. Lines 380-388: u_s is exact when the categories I_V^u are cofiltered; the description of I_V^u.
- The codirected (cofiltered) category conditions. Categories chapter, \label{categories-definition-codirected}. Lines 390-411: conditions (1)-(3) verified.
- The Zariski topology on a simplicial space. Same chapter, the definition of the site X_Zar. Lines 373-375: u sends the open U in X_n to the open f_n^{-1}(U) in Y_n.

Source reference. Original definition 04AZ: sources/stacks-chow/tags.txt, source lines 5634–5634. Prerequisite pointer invoked at proof line(s) 394; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00X5: sources/stacks-chow/tags.txt, source lines 1207–1207. Prerequisite pointer invoked at proof line(s) 381; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 00VC: sources/stacks-chow/tags.txt, source lines 1142–1142. Prerequisite pointer invoked at proof line(s) 383; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00X0: sources/stacks-chow/tags.txt, source lines 1202–1202. Prerequisite pointer invoked at proof line(s) 378; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of simplicial spaces. Then the functor $u : X_{Zar} \to Y_{Zar}$ which associates to the open $U \subset X_n$ the open $f_n^{-1}(U) \subset Y_n$ defines a morphism of sites $f_{Zar} : Y_{Zar} \to X_{Zar}$.*

Proof. It is clear that u is a continuous functor. Hence we obtain functors $f_{Zar,*} = u^s$ and $f_{Zar}^{-1} = u_s$, see Sites, Section 00X0. To see that we obtain a morphism of sites we have to show that u_s is exact. We will use Sites, Lemma 00X5 to see this. Let $V \subset Y_n$ be an open subset. The category $\mathcal{I}_V^u$ (see Sites, Section 00VC) consists of pairs (U, φ) where $\varphi : [m] \to [n]$ and $U \subset X_m$ open such that $Y(\varphi)(V) \subset f_m^{-1}(U)$. Moreover, a morphism $(U, \varphi) \to (U', \varphi')$ is given by a $\psi : [m'] \to [m]$ such that $X(\psi)(U) \subset U'$ and $\varphi \circ \psi = \varphi'$. It is our task to show that $\mathcal{I}_V^u$ is cofiltered.

We verify the conditions of Categories, Definition 04AZ. Condition (1) holds because $(X_n, \mathrm{id}_{[n]})$ is an object. Let (U, φ) be an object. The condition $Y(\varphi)(V) \subset f_m^{-1}(U)$ is equivalent to $V \subset f_n^{-1}(X(\varphi)^{-1}(U))$. Hence we obtain a morphism $(X(\varphi)^{-1}(U), \mathrm{id}_{[n]}) \to (U, \varphi)$ given by setting $\psi = \varphi$. Moreover, given a pair of objects of the form $(U, \mathrm{id}_{[n]})$ and $(U', \mathrm{id}_{[n]})$ we see there exists an object, namely $(U \cap U', \mathrm{id}_{[n]})$, which maps to both of them. Thus condition (2) holds. To verify condition (3) suppose given two morphisms $a, a' : (U, \varphi) \to (U', \varphi')$ given by $\psi, \psi' : [m'] \to [m]$. Then precomposing with the morphism $(X(\varphi)^{-1}(U), \mathrm{id}_{[n]}) \to (U, \varphi)$ given by φ equalizes a, a' because $\varphi \circ \psi = \varphi' = \varphi \circ \psi'$. This finishes the proof. $\square$